

Existing knowledge

Discussion with PMs

The screenshot displays the Oracle Enterprise Manager console interface. On the left, a sidebar lists various reports under the 'Resource Utilization report' category, including Heat Maps on CPU & IO, Consumption Details, Auto Scaling Events, System Failures or Outage, CPU I/O, Speed, Disk, and AMPs, Nodes, Cluster & Environment. The main content area is titled 'Query/Database Performance report' for 'This Month'. It provides a summary of query performance, stating that the system is performing well with a 'Good' guard chart. Below this, there are two charts: 'Total Queries' broken down by status (Good, Bad, Suspended) and 'Breakdown by Workload Type' (OLTP, DSS, etc.). The right sidebar, titled 'Workshop Agenda', lists several reports: Query Monitoring module in Lake Console, Query Performance report, Resource Utilization report, and Insights Total Cost of Ownership, each with a brief description of its content.

Query Monitoring Module in Lake Console

Issue	Configuration	Impact
High CPU usage	Excessive parallelism or large batch sizes	Increased costs, slower processing
Memory pressure	Large data volumes or complex queries	Out of memory errors, task failures
Slow query execution	Unoptimized queries, lack of partitions	Increased latency, higher costs
Resource contention	Overloading the cluster with too many tasks	Queueing delays, inconsistent performance
Network bottlenecks	High data transfer volumes between nodes	Reduced throughput, increased latency
Configuration errors	Incorrect settings for parallelism, memory, or timeouts	Task failures, resource wastage
Hardware issues	Faulty nodes or insufficient resources	Task failures, reduced availability
Software bugs	Issues with the query engine or Lakehouse components	Unpredictable behavior, crashes
Security concerns	Unauthorized access or data leakage	Compliance violations, data loss
Operational challenges	Lack of monitoring and alerting	Delayed detection of issues, manual intervention

Query monitoring page shows the top 100 longest queries by runtime or duration for the selected time window

Bad Queries in the last month

- Long Running
- CPU
- I/O, Disk, Spill
- Blocking
- Run for 10min then got cancelled

Query Monitoring

Summary View

Global Bad ABC SQL Duration Query ID User Status Compile Group Workload Classification

Query Details

Highlight why it's bad!

CPU Usage: 100% (too high for this metric)

IO: 100MB/s (too high)

Spill: 100MB (too high)

Blocking: blocked by

Query Explain

Table 1 Table 2 Table 3

File 1 File 2 File 3

Query

SQL: SELECT * FROM table1

Query ID: 123456789

Query Status: Running

Query Type: SELECT

Query User: root

Query Group: root

Existing knowledge - Statuses & Filters

[illegible]

What and Why of query summary

What and Why of query summary

User Need

- User would like to see what RAID has happened for ALL Queries.
- User might have some preference on what queries he wants to see.
- User would like to see all the queries going on currently and which is taking more CPU/I/O, Space etc.
- Day-to-day life of DBA might have currently running queries only instead of All Queries.

User Need/ Designer Goals

Without doing any actions/least amount of action user should be able to see the cookies that are of most impact.

Improve

- List of all status + description
- List of all metrics + description
- List of all metrics + description